BME 3030: Effects of Mental Exertion on Heart Rate

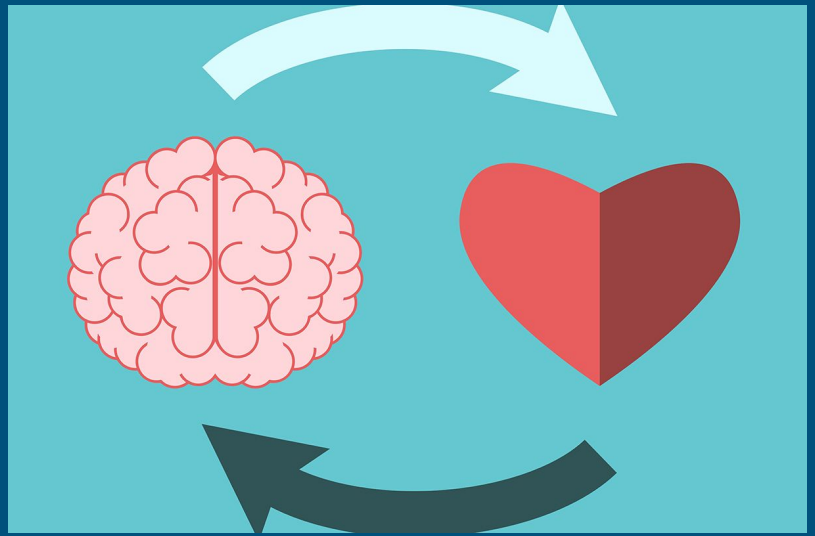
Antranig, Klaus

Hypothesis

During periods of mental exertions, a person's heart rate will be greater than during periods without mental exertion.

Justification:

- Acute psychological stress is associated with increased heart rate (Cook, et al.)
- During tests, personally observed increased heart rate, especially when short amounts of time remained



Experimental Design

Control:

- 3 Minutes, copying numbers down onto iPad
- Accuracy not graded
- No prizes

Experiment:

- 3 Minutes, solving multiplication problems on iPad
- Accuracy graded
  - Penalized for wrong and blank answers
- Possibility to win candy if obtain certain accuracy with minimum number of questions solved
- Timer displayed

n = 11

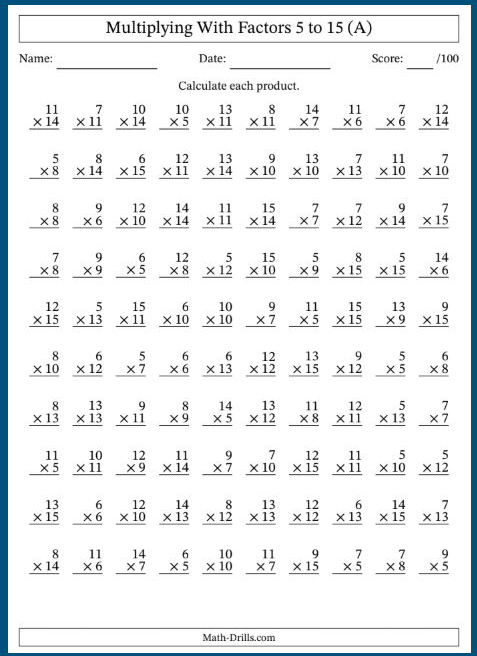
Experimental Design Cont.

Control:

Line					
101	19223	95034	05756	28713	96409
102	73676	47150	99400	01927	27754
103	45467	71709	77558	00095	32863
104	52711	38889	93074	60227	40011
105	95592	94007	69971	91481	60779
106	68417	35013	15529	72765	85089
107	82739	57890	20807	47511	81676
108	60940	72024	17868	24943	61790
109	36009	19365	15412	39638	85453
110	38448	48789	18338	24697	39364
111	81486	69487	60513	09297	00412
112	59636	88804	04634	71197	19352
113	62568	70206	40325	03699	71080
114	45149	32992	75730	66280	03819
115	61041	77684	94322	24709	73698
116	14459	26056	31424	80371	65103
117	38167	98532	62183	70632	23417
118	73190	32533	04470	29669	84407
119	95857	07118	87664	92099	58806
120	35476	55972	39421	65850	04266
121	71487	09984	29077	14863	61683
122	13873	81598	95052	90908	73592
123	54580	81507	27102	56027	55892
124	71035	09001	43367	49497	72719
125	96746	12149	37823	71868	18442
126	96927	19931	36809	74192	77567
127	43909	99477	25330	64359	40085

Experiment:

- 35 questions answered with >= 80%, 1 prize
- 50 questions answered with >= 80%, 2 prizes
- -1 every wrong answer
- +0 every blank answer
- +1 every right answer



Multiplying With Factors 5 to 15 (A)

Name: _____ Date: _____ Score: ____ /100

Calculate each product.

11 × 14	7 × 11	10 × 14	10 × 5	13 × 11	8 × 11	14 × 7	11 × 6	7 × 6	12 × 14
5 × 8	8 × 14	6 × 15	12 × 11	13 × 14	9 × 10	13 × 10	7 × 13	11 × 10	7 × 10
8 × 8	9 × 6	12 × 10	14 × 14	11 × 11	15 × 14	7 × 7	7 × 12	9 × 14	7 × 15
7 × 8	9 × 9	6 × 5	12 × 8	5 × 12	15 × 10	5 × 9	8 × 15	5 × 15	14 × 6
12 × 15	5 × 13	15 × 11	6 × 10	10 × 10	9 × 7	11 × 5	15 × 15	13 × 9	9 × 15
8 × 10	6 × 12	5 × 7	6 × 6	6 × 13	12 × 12	13 × 15	9 × 12	5 × 5	6 × 8
8 × 13	13 × 13	9 × 11	8 × 9	14 × 5	13 × 12	11 × 8	12 × 11	5 × 13	7 × 7
11 × 5	10 × 11	12 × 9	11 × 14	9 × 7	7 × 10	12 × 15	11 × 11	5 × 10	5 × 12
13 × 15	6 × 6	12 × 10	14 × 13	8 × 12	13 × 13	12 × 12	6 × 13	14 × 15	7 × 13
8 × 14	11 × 6	14 × 7	6 × 5	10 × 10	11 × 7	9 × 15	7 × 5	7 × 8	9 × 5

Math-Drills.com

Circuit Design

INA128

Gain = 167

Right Abdomen Drive:

Gain = 10-100

Low Pass Filter:

f_c @ 40 Hz

High Pass Filter 1:

f_c @ 0.1 Hz

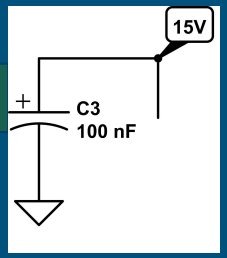
DAC Op Amp:

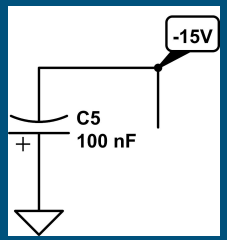
G = 69.75

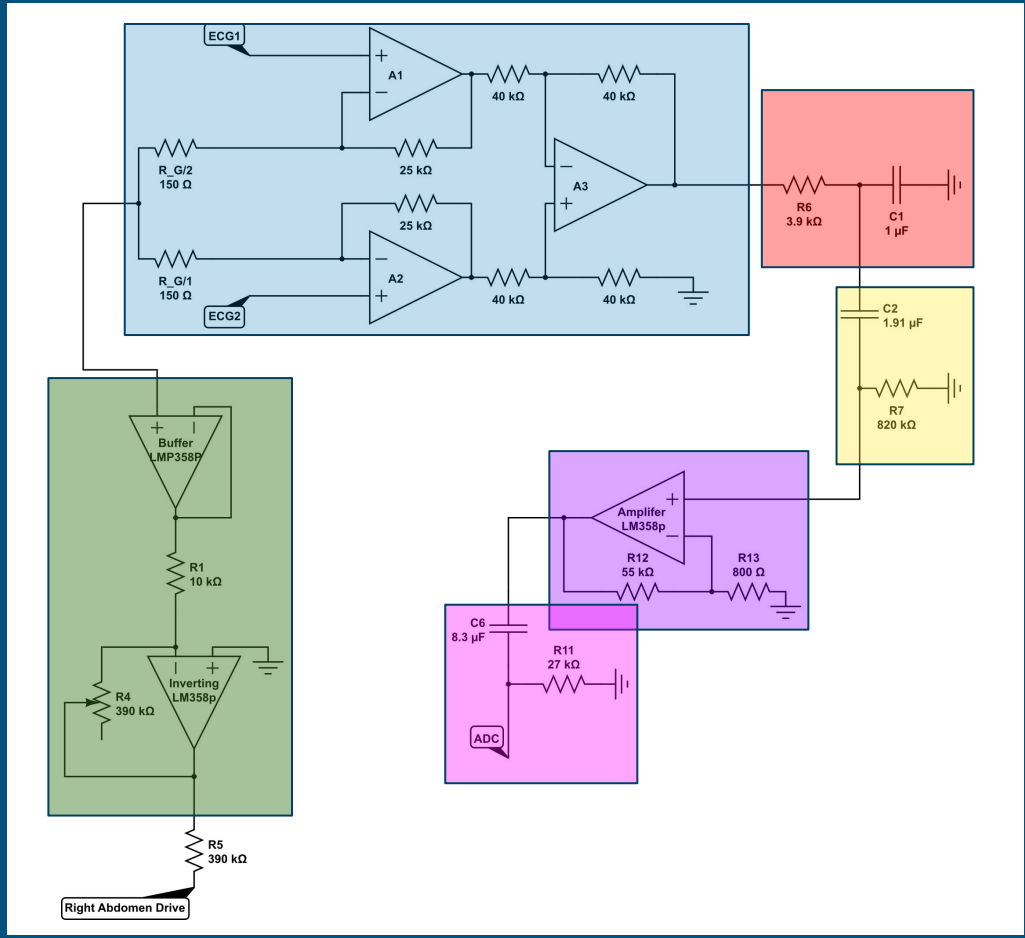
High Pass Filter 2:

f_c @ 0.71 Hz

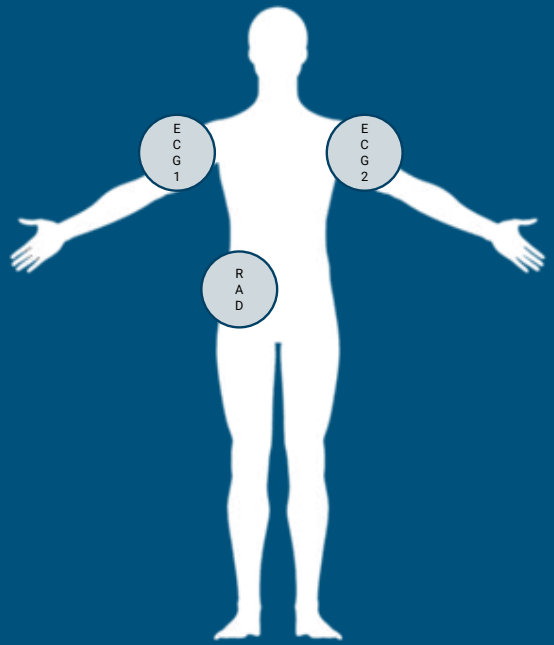
Bypass Capacitors:

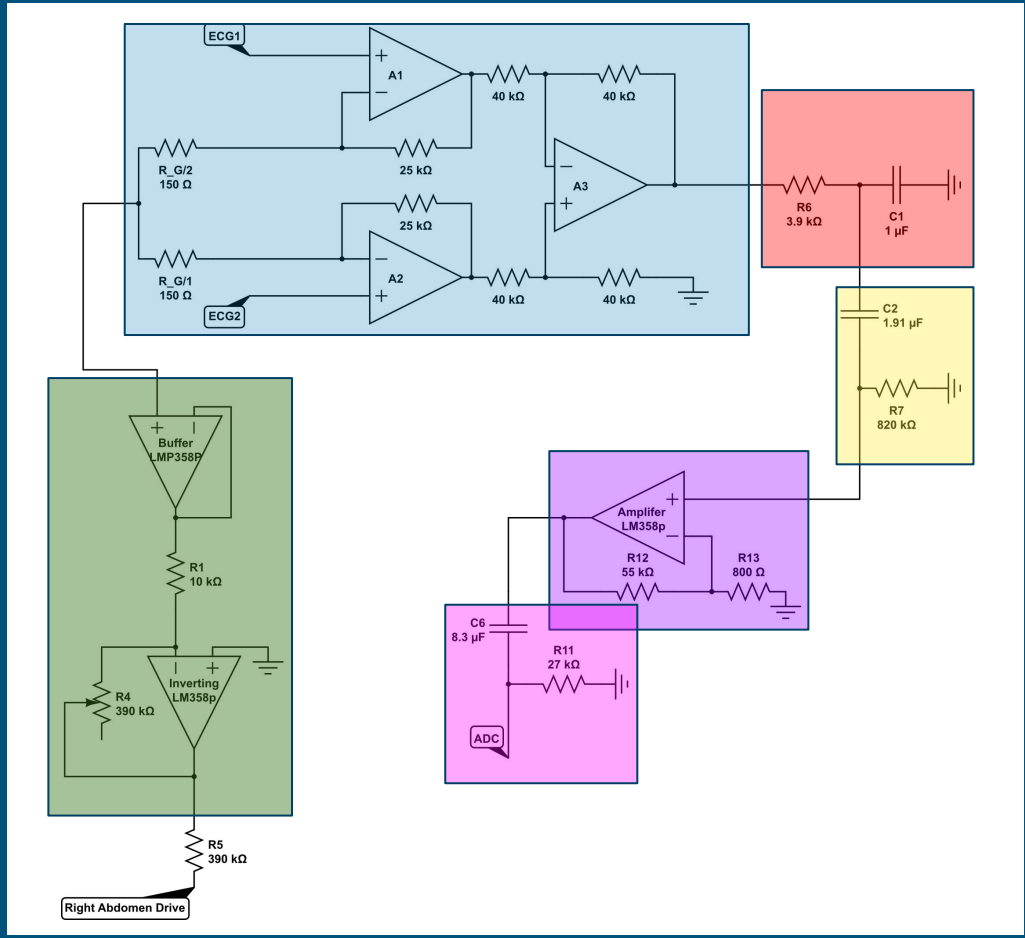




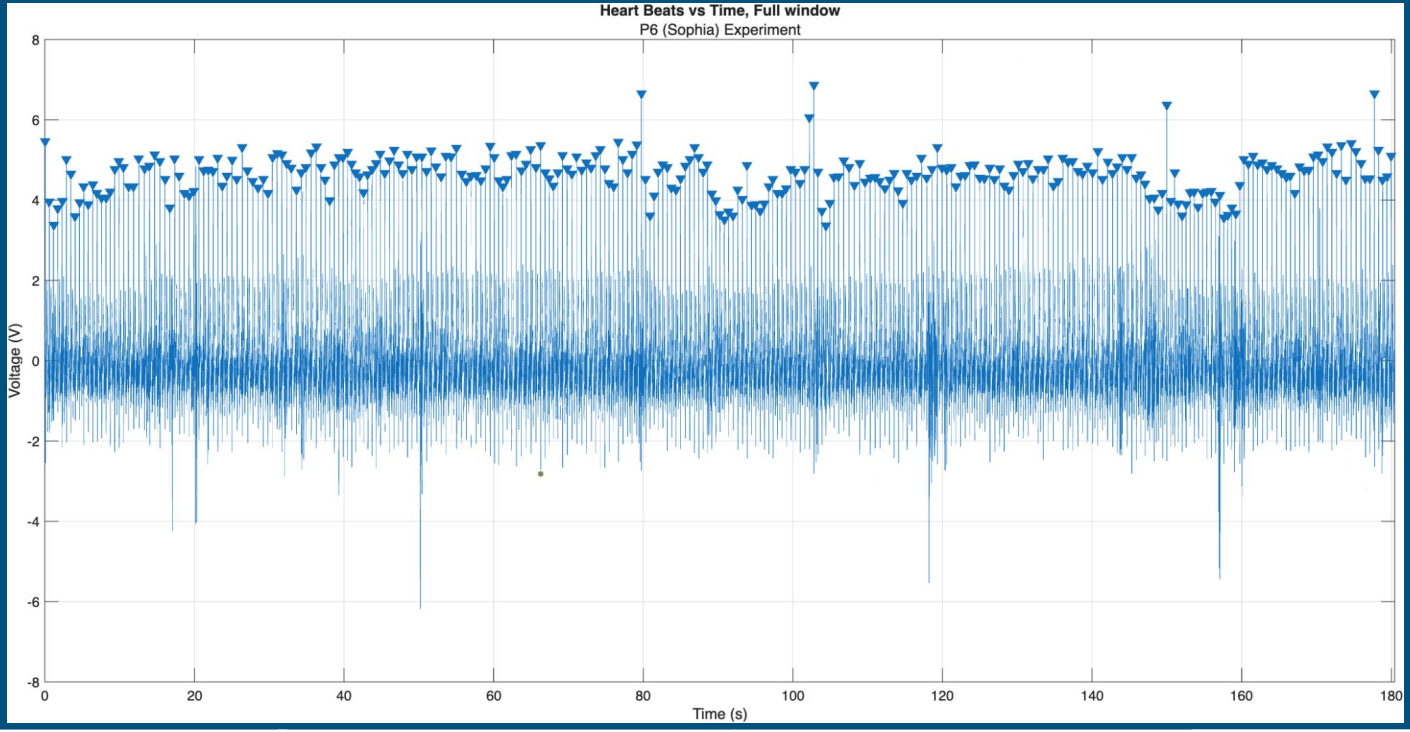


Circuit Design





Cleaning the Signal: Raw Signal Output



Import CSV

Compare filtered & unfiltered signals

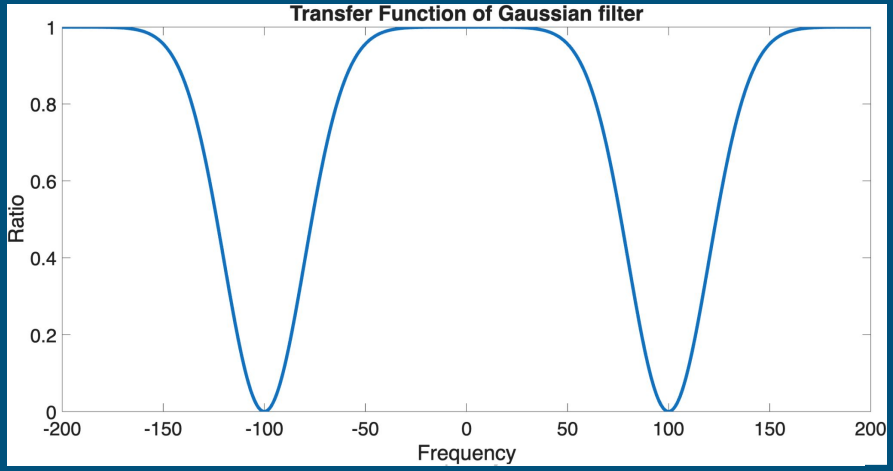
Compare Instantaneous Heart Rates from Various FindPeaks

Determine if Further Conditioning is Needed

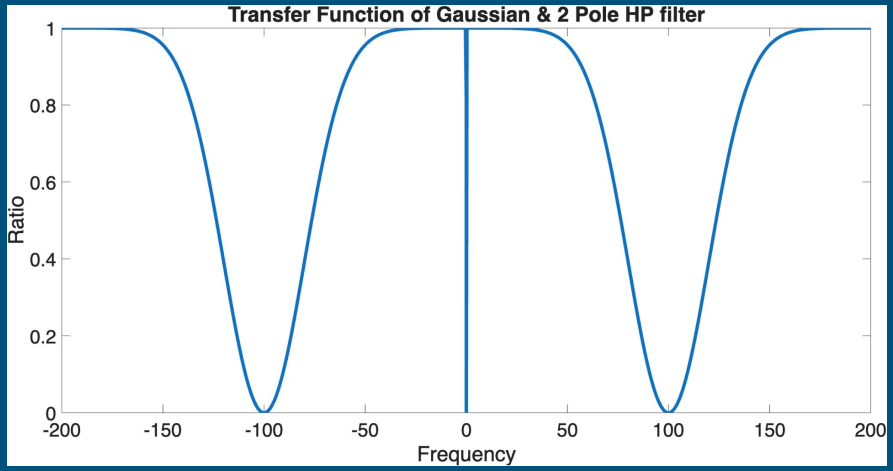
Export & Sanity Check

Cleaning the Signal: Workflow

Transfer functions of the 2 filters used:



Gaussian



Gaussian & 2 Pole HPF

Import CSV

Compare filtered & unfiltered signals

Compare Instantaneous Heart Rates from Various FindPeaks

Determine if Further Conditioning is Needed

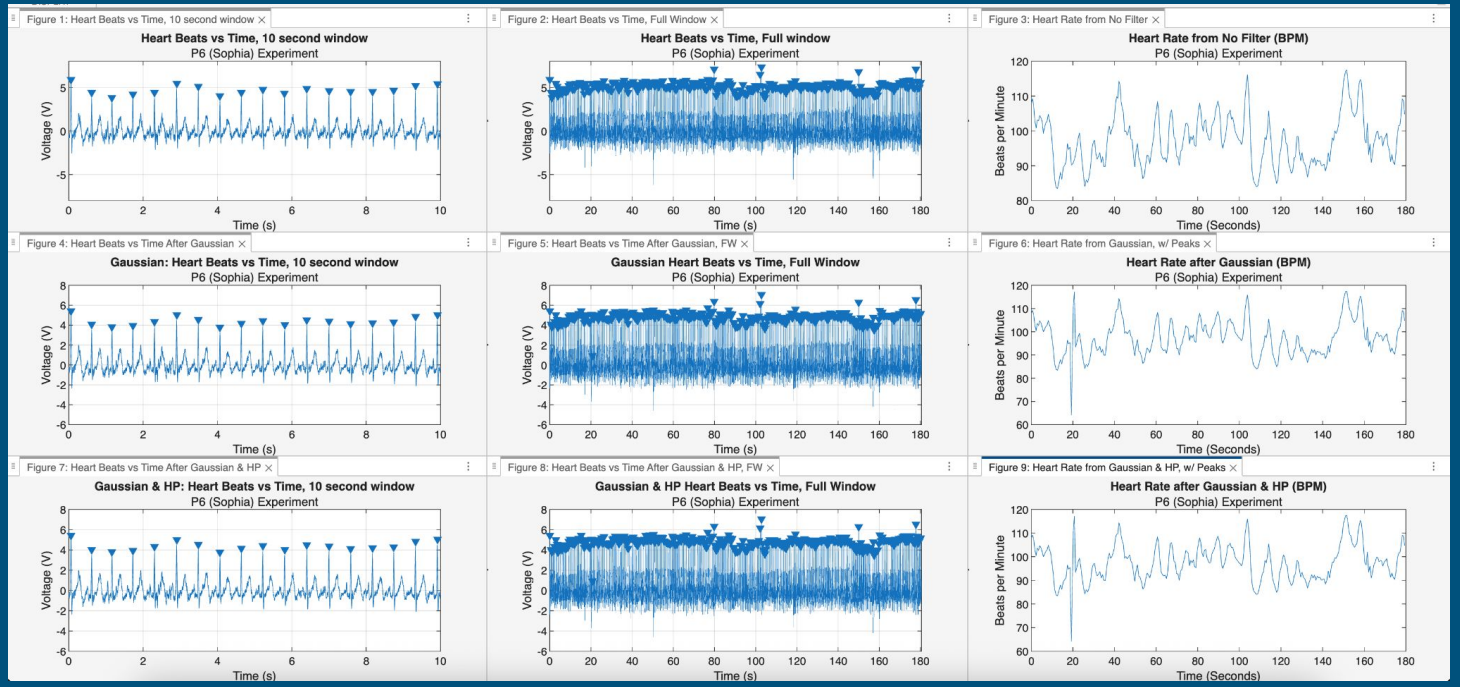
Export & Sanity Check

Cleaning the Signal: Workflow

No Filter →

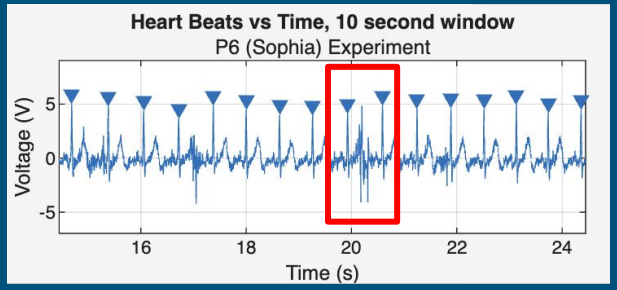
Gaussian →

Gaussian & HPF →

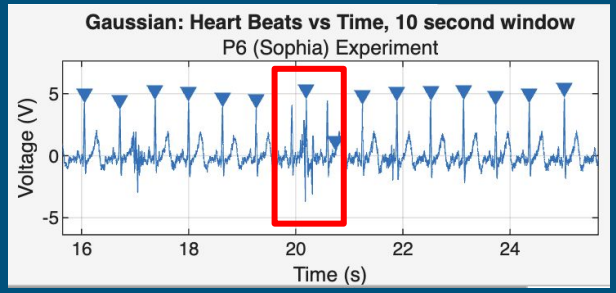


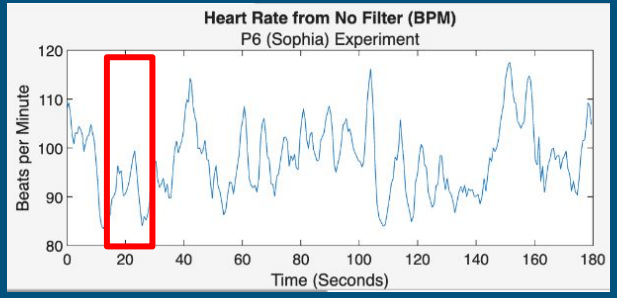
Import CSV → Compare filtered & unfiltered signals → Compare Instantaneous Heart Rates from Various FindPeaks → Determine if Further Conditioning is Needed → Export & Sanity Check

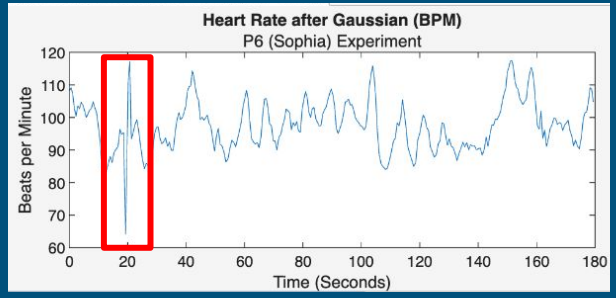
Cleaning the Signal: Workflow



Here, no filter was best.

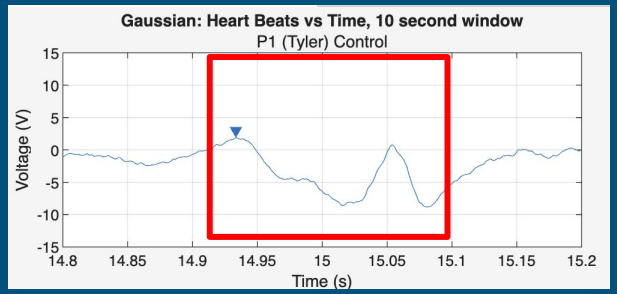


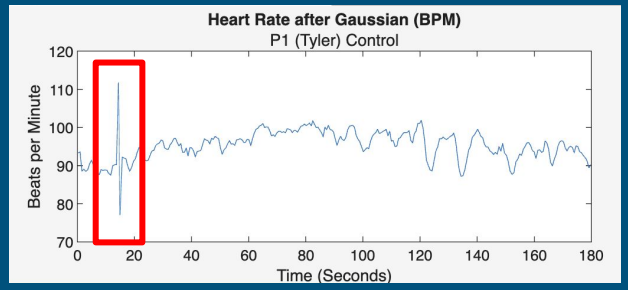




Import CSV → Compare filtered & unfiltered signals → Compare Instantaneous Heart Rates from Various FindPeaks → Determine if Further Conditioning is Needed → Export & Sanity Check

Cleaning the Signal: Workflow



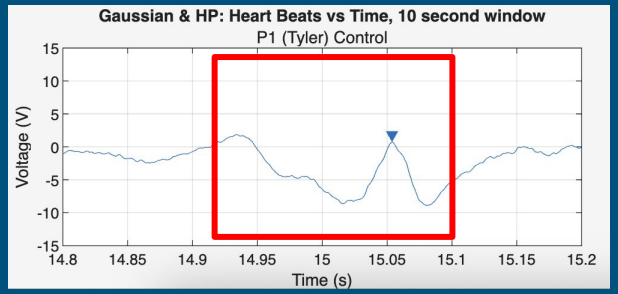


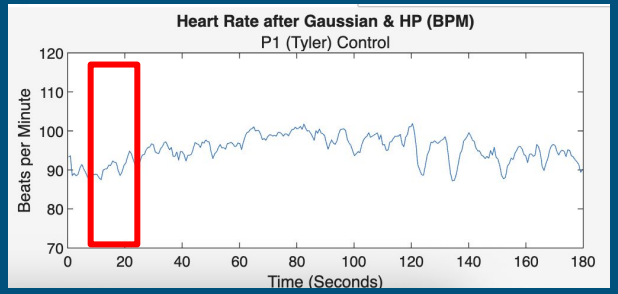
Here, FindPeaks criteria needed to be refined:

- Increasing peak prominences
- Setting: minpeakwidth.

Other Possible Changes:

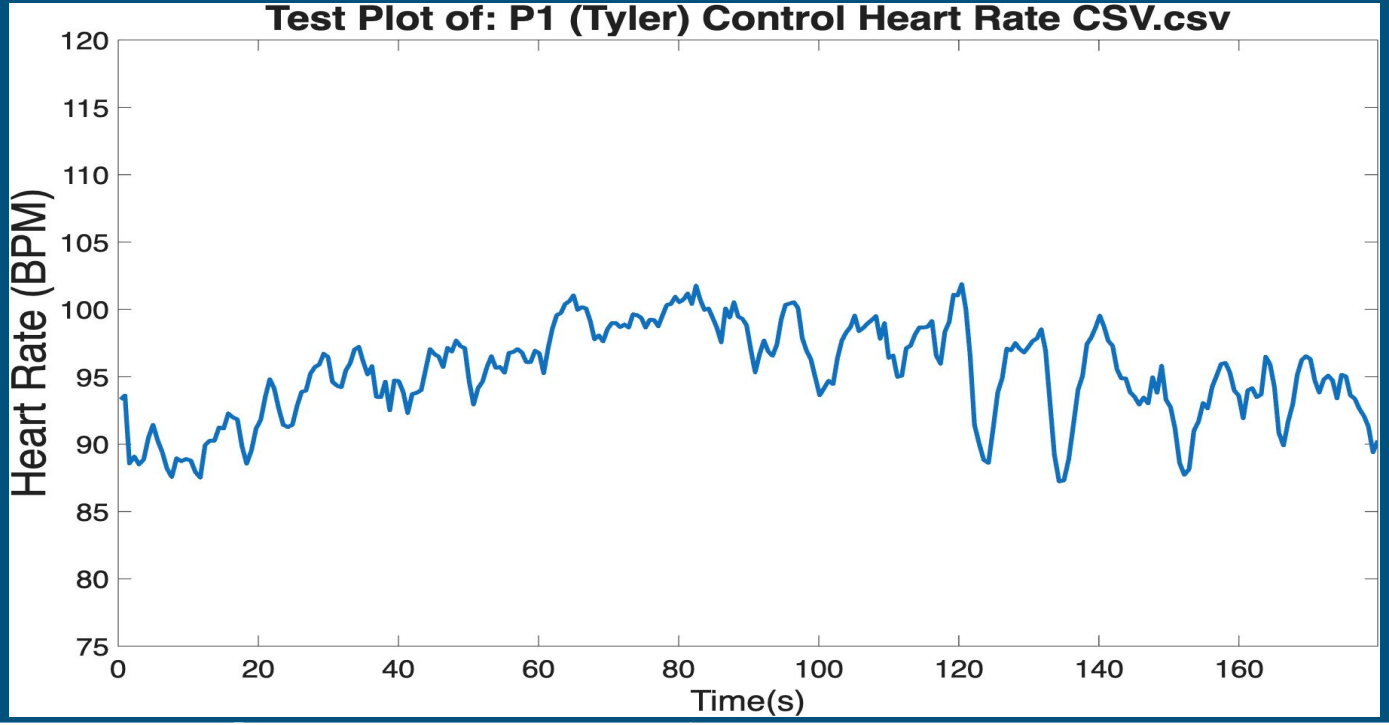
- Alter Filters
- Specific Windows of Data





Import CSV → Compare filtered & unfiltered signals → Compare Instantaneous Heart Rates from Various FindPeaks → Determine if Further Conditioning is Needed → Export & Sanity Check

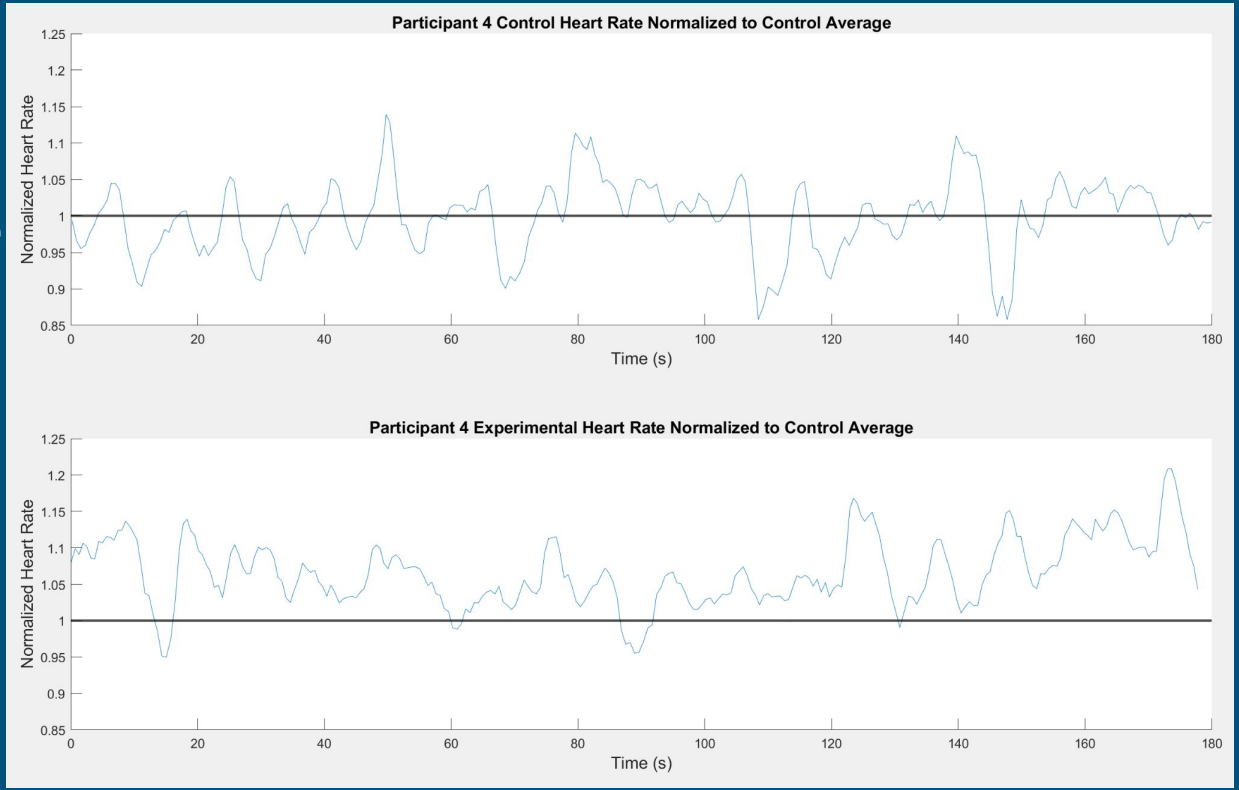
Cleaning the Signal: Workflow



Import CSV → Compare filtered & unfiltered signals → Compare Instantaneous Heart Rates from Various FindPeaks → Determine if Further Conditioning is Needed → Export & Sanity Check

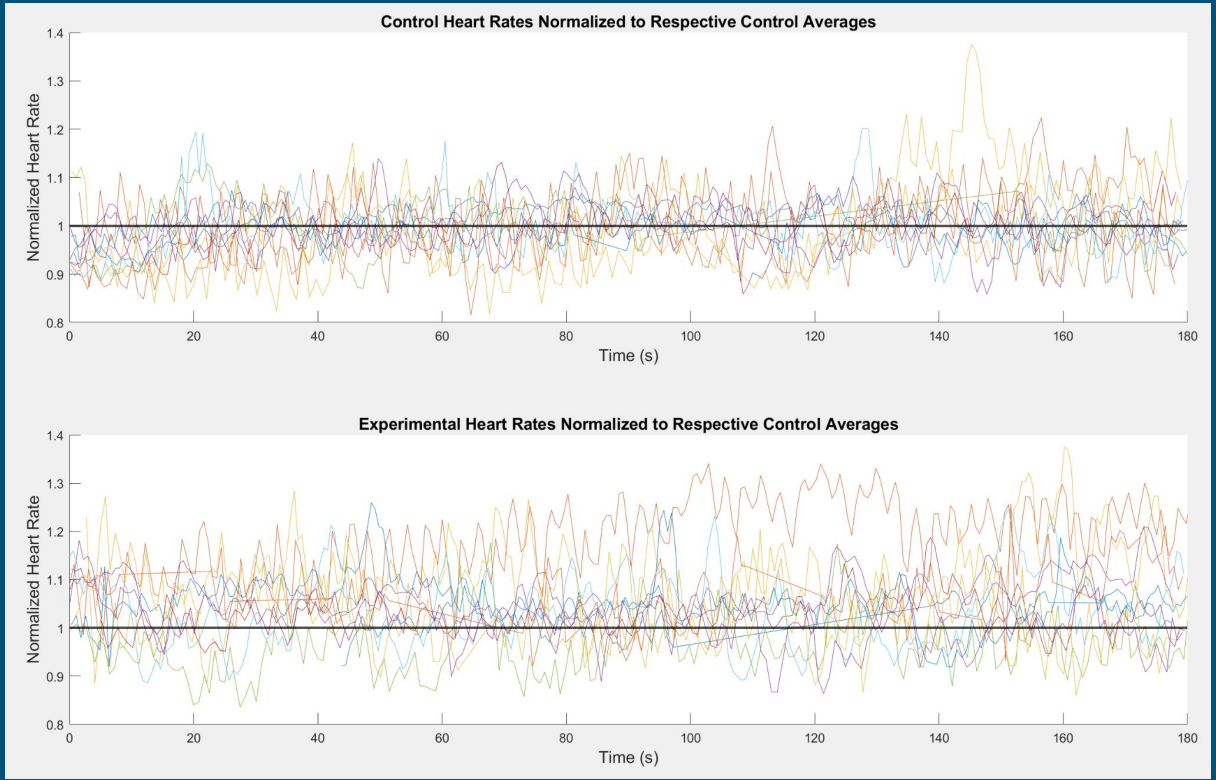
Raw Data Normalization

- Each participant's data is **normalized to the 3-minute average of their heart rate in the control case**
- The bold black line is **y=1**, showing where a given data point lies with respect to the control average



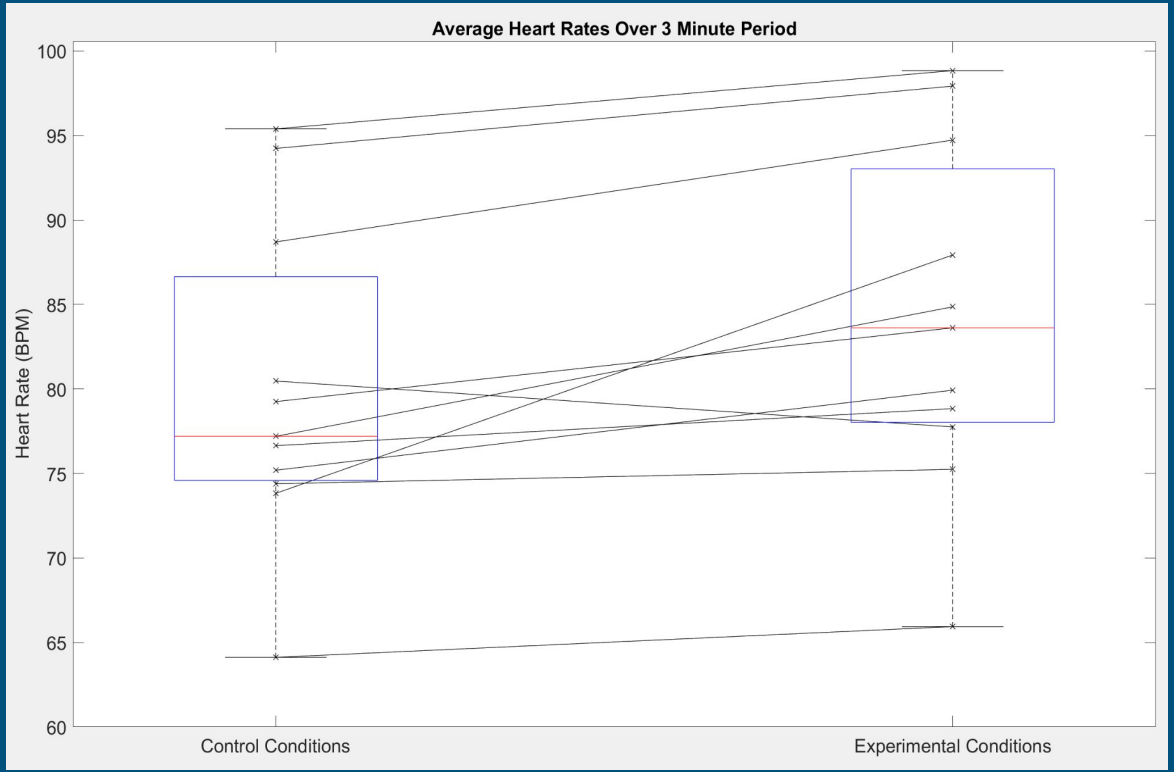
Data Summary

- Experimental heart rate values tend to be above the baseline much more often than control values **throughout the entire time frame**
- We will test for a difference in average heart rate across the full interval



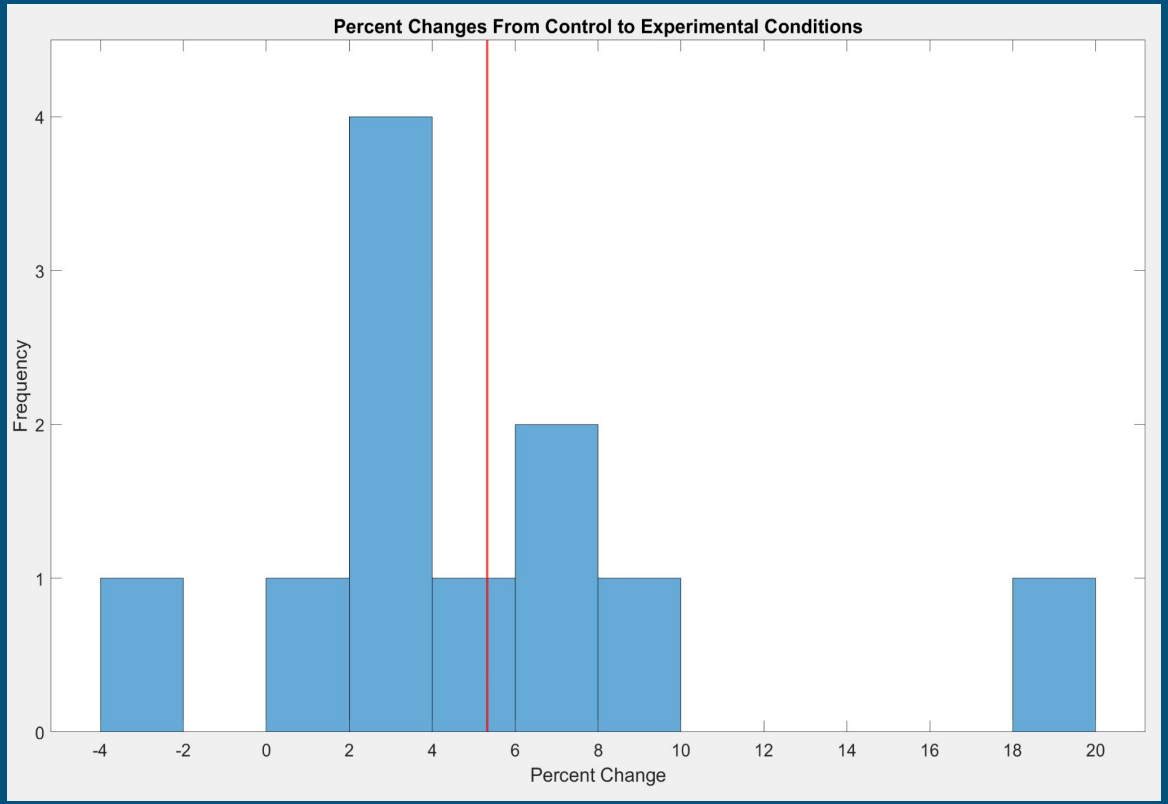
Data Summary

- Increase in average heart rate observed from control conditions to experimental conditions for almost all participants



Percent Changes

- Red line is **average percentage increase in heart rate from control to experimental conditions**
- Average increase: 5.33%



Statistical Significance: Heart Rate Averages

- Let μ_{copy} be the average of heart rate in the control condition of copying numbers and μ_{calc} be the average of heart rate in the experimental condition of performing calculations.

  Then:

$$H_0: \mu_{calc} - \mu_{copy} = 0$$
$$H_a: \mu_{calc} - \mu_{copy} > 0$$

- Performing a right-tailed paired t test with $\alpha=0.05$ on the slopes results in a p value of **p=0.0043**, meaning we **reject the null hypothesis.**
- This provides strong evidence that a person's average heart rate increases during mental exertion, supporting our original hypothesis

Limitations and Considerations

- Circuit Design:
  - When using a right-abdomen drive, we found that with certain participants the circuit was sensitive to muscle contractions from speaking or moving
  - Moving the drive ECG to the right leg may have helped alleviate this
- Experimental Design:
  - We did not precisely control the amount of time between the control and experimental trials for each participant.
  - It is possible that the control writing may have elevated a participant's heart rate over time, and this elevation bled through into the calculation trials.
    - Low overall HR increase in control trials indicate that this was likely not significant, but it should have been controlled more precisely (uniform time between trials, etc)